

Lightweight Mobile CNNs for Point-of-Care Skin Lesion Screening

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Problem

Enabling on-device skin lesion triage in low-resource settings with limited compute.

Method

A MobileNetV3-based architecture optimized for latency and model size.

Dataset

A subset of the ISIC dataset.

Evaluation

Accuracy versus inference-time trade-off on mobile hardware.

Results

The model achieved 87% accuracy at 45ms inference time on a mid-range smartphone CPU.

Limitations

Not validated on smartphone camera images captured under real field conditions.